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Summary

Backend and platform software engineer, sole active maintainer of a Django/DRF grading platform serving 1,950+ users on a 4-VM container fleet — including its sandboxed multi-language execution engine and per-token cost-attribution layer. Depth in concurrency, query performance, container isolation, and application security, behind 1,100+ automated tests and gated CI/CD.

Experience

Rutgers University, Application Developer (Software Engineer)

June 2025 - Present

- Sole active engineer on codePost, a Django/DRF grading platform serving 1,950+ users — 1,828 students, 57 graders, 17 course admins — across 19 courses and 74 assignments; 80K lines of Python, 91 schema migrations, and 30+ REST resource ViewSets running as 12 containers on a 4-VM Docker Compose fleet behind nginx and gunicorn/Uvicorn ASGI workers.
- Built the sandboxed execution engine: container-per-run Docker isolation for seven language runtimes, each with a paired per-cell Jupyter notebook executor, capped at 300s wall time, 1 GiB memory with swap disabled, one CPU, 500 PIDs, no network, and all Linux capabilities dropped.
- Cut redundant grading compute with a SHA-256 content-hash execution cache that skips the container run entirely on a hit, backed by an index built for the (file, most-recent-run) query shape, plus test-suite batching that runs each category script once instead of once per test case.
- Removed N+1 query storms from the permission chain with a purpose-built per-request role-memoization layer and 77 eager-loading sites; added 17 composite indexes, F()-expression race-safe counters, and compare-and-set idempotency guards on concurrent AI jobs.
- Built per-token AI cost attribution: a USD rate table across two providers with cached-token discount modeling, charges quantized to one-millionth of a dollar in a Decimal(10,6) column, three-level course-over-organization-over-default rate-override precedence, and hourly, daily, and monthly rollups at course, organization, and platform scope.
- Hardened operations with a five-probe health endpoint — database round-trip latency, live Celery worker count, cache write and read-back, pending migrations, disk headroom — self-healing autograder environments that promote above an 0.8 success rate and auto-roll-back below 0.5, and 992 backend tests running in parallel behind a four-job CI gate.
- Took sole ownership of a 5,200-commit legacy React SPA — 103K hand-written TS/TSX, 364 components, 167 routes — migrating Create React App to Vite and React Router v5 to v7 in one hop across 398 files, adding 39 lazy-loaded route boundaries, and gating deploys on a strict type check plus 173 tests.
- Contributed 337 commits to a shared 229-role Ansible fleet repository, rebuilding JupyterHub with four pluggable auth backends (Kerberos, CAS, Azure AD, local) and per-user systemd cgroup isolation: 40 GB memory ceiling, swap off, OOM-kill on pressure, 10 GB quota, and an 8-hour CPU-time killer.
- Designed and published jupyter-assignments to npm and PyPI: a JupyterLab extension with an abstract platform adapter (13 methods, three backends) and a four-tier role hierarchy enforced by a decorator at the platform layer, covered by 320 Python tests.

Rutgers University, Student Lab Technician

January 2023 - May 2025

- Managed 10 Hackerspace workspaces, writing student-focused documentation and maintaining the WordPress site, modernizing the space.
- Built a Raspberry Pi computer-vision learning station with MediaPipe, used for walk-in student workshops.
- Built a 3D printer monitoring service (Express and React) streaming live video of the printer bay so students could check job status without walking over.

Swish, Full-Stack Developer

July 2021 - December 2021

- Automated account creation on web platforms using Puppeteer and Node.js, with Mocha-based regression tests covering the signup flows.
- Migrated the CLI to an Electron desktop application, replacing flag-based invocation with a GUI workflow.

C-Tech, Computer Technician

July 2019 - December 2021

- Automated computer initialization with Powershell scripts, cutting build time from 40 to 20 minutes.
- Implemented a secure HDD erasure workflow processing dozens of drives daily, onboarding 3 new enterprise clients.

Skills

Languages: Python, TypeScript, JavaScript, C, Java, GoLang, SQL, PowerShell, Bash

Frameworks: Django, Django REST Framework, Django Channels, Celery, React, Next.js, Flask, Express, JupyterLab, Vite, Electron

Data: MariaDB / InnoDB, MongoDB, Redis, LevelDB, Django ORM, Schema Design and Migrations, Query Optimization and Indexing

Infrastructure: Docker, Docker Compose, Ansible, Nginx, gunicorn / Uvicorn (ASGI), Linux, systemd, NFS, Proxmox, VMware, OpenStack, AWS, GitHub Actions, GitLab CI, Zabbix

Architecture: REST and OpenAPI Design, Distributed Task Queues, Caching and Invalidation, Container Isolation, Idempotency and Race Safety, RBAC and Authorization, Rate Limiting, Usage Metering and Cost Attribution, Health Checks and Self-Healing, CI/CD and Staged Deploys, WCAG 2.1 AA

Security: Application Security, OWASP Top 10, Authorization and Access Control, CAS / SAML / Kerberos SSO, JWT and API Key Management, Encryption at Rest, Suricata, pfSense

Testing: pytest, pytest-xdist, Vitest, Testing Library, Playwright, axe-core, Molecule, Mocha

Other: Git, Ollama, OpenAI and Gemini APIs, Tree-sitter, WebContainers, OpenCV, MediaPipe

Projects

CompLock

November 2023 - Present

- Developed SSH command and control software for networked computers in TypeScript, leveraging LevelDB and SSH2, utilized by RUSEC in CCDC competitions.
- Reduced password rotation on 30 machines from 5 minutes to 30 seconds with automated testing via GitHub Actions and Mocha.

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April 2025 - Present

- Designed and built a Next.js 16 / React 19 portfolio and blog with a markdown publishing pipeline, PWA support, server-resolved theming, and a keyboard command palette.
- Engineered an in-browser code playground with WebContainers that boots a Node.js runtime client-side to run my published ts-declaration-json package live, with no server execution.
- Implemented a simulated Linux shell in TypeScript (custom filesystem, user system, and coreutils) served as the site's 404 page, plus Spotify OAuth with token refresh and an AI task-planning agent with tool calling.

AppTracker

2025

- Built a TypeScript application that tracks job applications by reading incoming emails via IMAP, classifying them with a Bayesian classifier, and generating status updates using Ollama AI models.
- Integrated Discord bot notifications and MongoDB storage to provide real-time application status updates.

Education

Bachelor of Arts, Computer Science

May 2025

- Rutgers University — Coursework: Computer Security, Software Methodology, Computer Architecture.

Leadership

RUSecurity (Student Cybersecurity Club), Vice President

January 2023 - May 2025

- Directed network administration for CCDC, with the team placing 4th in 2024.
- Led club infrastructure setup on Proxmox and VMware ESXi, standing up the practice range the team trained on.
- Led development of the club's BlackBox machine environment with Wazuh detection rules and the ToolBox hardening toolkit used for CCDC preparation.